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Certificate of participation

Issued to : Filip-Alexandru Mocanca

Title of the course :

Module 5 - Part 2 : DevOps and Pipeline Security

Date of the course : from 21.04.2026 to 27.04.2026

Hours of training : 14 (100 %)

Attendance rate : 100 %

Esch-sur-Alzette: 29 April 2026

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Module 5 - Part 2 : DevOps and Pipeline Security

Content:

Introduction and Environment Setup

DevSecOps Overview and Key Concepts

• Introduction to DevSecOps and its importance in modern development pipelines

• DevOps vs DevSecOps: The shift-left approach

• Security as Code: Embedding security throughout the CI/CD pipeline

• Compliance, governance, and security controls

CI/CD Pipeline Setup

Hands-on setup of a CI/CD pipeline using Jenkins/GitLab CI/ArgoCD:

• Install and configure Jenkins/GitLab/ArgoCD
• Build a basic pipeline
• Integrating version control (Git)

Secure CI/CD practices:

• Hardening Jenkins and GitLab CI
• Secrets management with Vault/KMS
• Configuring role-based access control (RBAC)

Secure Build and Deployment

Implement static code analysis tools (SonarQube, OWASP):

• Integrating with the pipeline for security analysis
• Custom rules to detect security misconfigurations

Container security scanning:

• Using Trivy to scan Docker images for vulnerabilities
• Automating scans as part of the pipeline

Messaging Queue (Kafka/RabbitMQ) Security and Integration

Hands-on setup of Kafka or RabbitMQ:

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• Deploy Kafka/RabbitMQ in a Kubernetes cluster
• Securing Kafka brokers, Zookeeper, and communications (TLS, ACLs)
Misconfigurations in message queues:

• Identifying common Kafka/RabbitMQ security risks (e.g., open brokers, no encryption)
• Using Kafka to send/receive logs for monitoring
Integrating Kafka with Grafana for monitoring:

• Setting up a dashboard to visualize Kafka metrics
• Identifying Kafka-related security incidents via Grafana

Advanced Topics and Hands-on Labs

Infrastructure as Code (IaC) Security

Introduction to Terraform/Ansible/Pulumi:

• How IaC fits into the DevSecOps pipeline
• Writing and deploying secure infrastructure code
Hands-on demo:

• Deploying infrastructure using Terraform
• Validating security posture with tools like Checkov, TFLint, or Terrascan

CI/CD pipeline integration for IaC security checks

Runtime Security

Security at runtime with Kubernetes:

• Implementing network policies, pod security policies, and RBAC in Kubernetes
• Best practices for securing Kubernetes deployments
• Monitoring runtime security using Falco
Hands-on demo:

• Deploying and securing Kubernetes workloads with Pod Security Standards (PSS)
• Detecting security violations and misconfigurations in real-time with Falco

Incident Detection and Response

Introduction to monitoring and alerting in DevSecOps:

• Setting up monitoring tools (Grafana, Prometheus)

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• Alerting with Alertmanager and PagerDuty
Detecting and responding to incidents:

• Configuring security alerts for real-time threat detection
• Correlating Kafka logs with security incidents
• Incident response workflow in a DevSecOps pipeline
• Hands-on lab: Set up a secure monitoring and alerting solution

Advanced DevSecOps Best Practices

Hardening and securing microservices:

• Service Mesh (Istio/Linkerd) for security and observability
• Enforcing zero-trust policies across services
• Managing secrets securely (HashiCorp Vault)
Misconfiguration detection using Grafana:

• Setting up real-time dashboards to visualize security misconfigurations
• Automated alerts for detecting configuration drifts
Hands-on lab:

• Simulating security misconfigurations in Kafka/RabbitMQ and Kubernetes
• Detecting and visualizing issues with Grafana dashboards

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Learning Outcomes:

• Build and secure CI/CD pipelines with static code analysis and container scanning.
• Secure message queues (Kafka/RabbitMQ) and monitor them using Grafana.
• Implement infrastructure as code (IaC) security using Terraform and automate checks.
• Harden Kubernetes deployments and detect runtime security issues.
• Set up incident detection, monitoring, and alerting systems for proactive security.

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